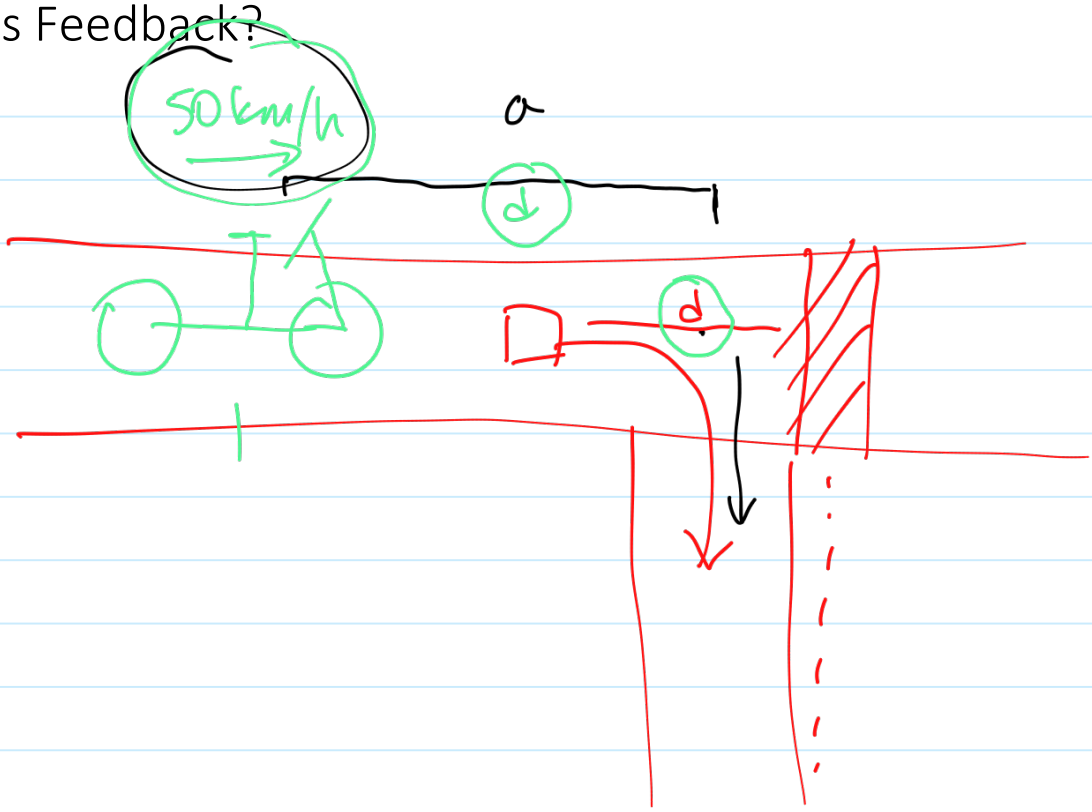
Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

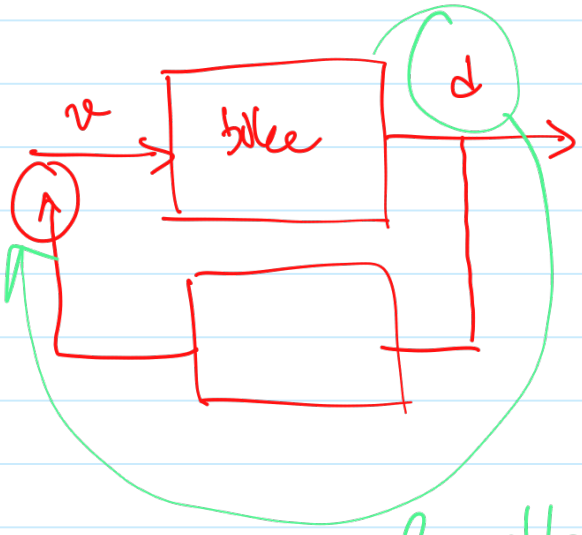
- Chapter 11.1 The General Feedback Structure
- Chapter 11.2 Some Properties of Negative Feedback

What is Feedback?



Without feedback

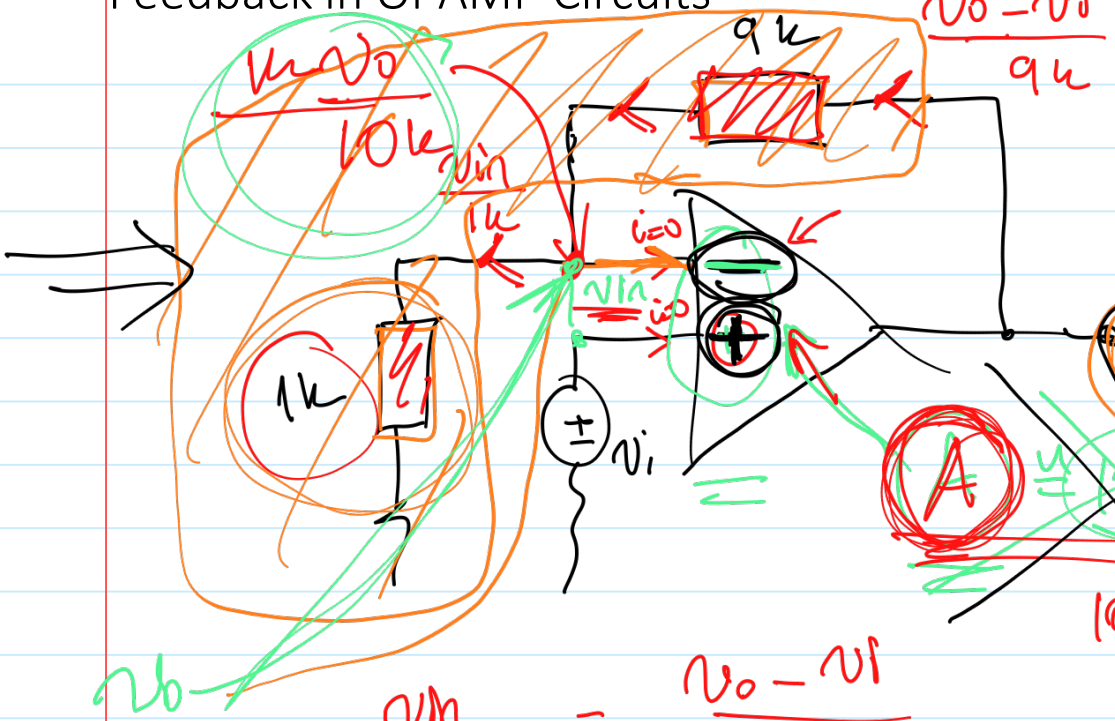
$\vec{v}$, d



feedback

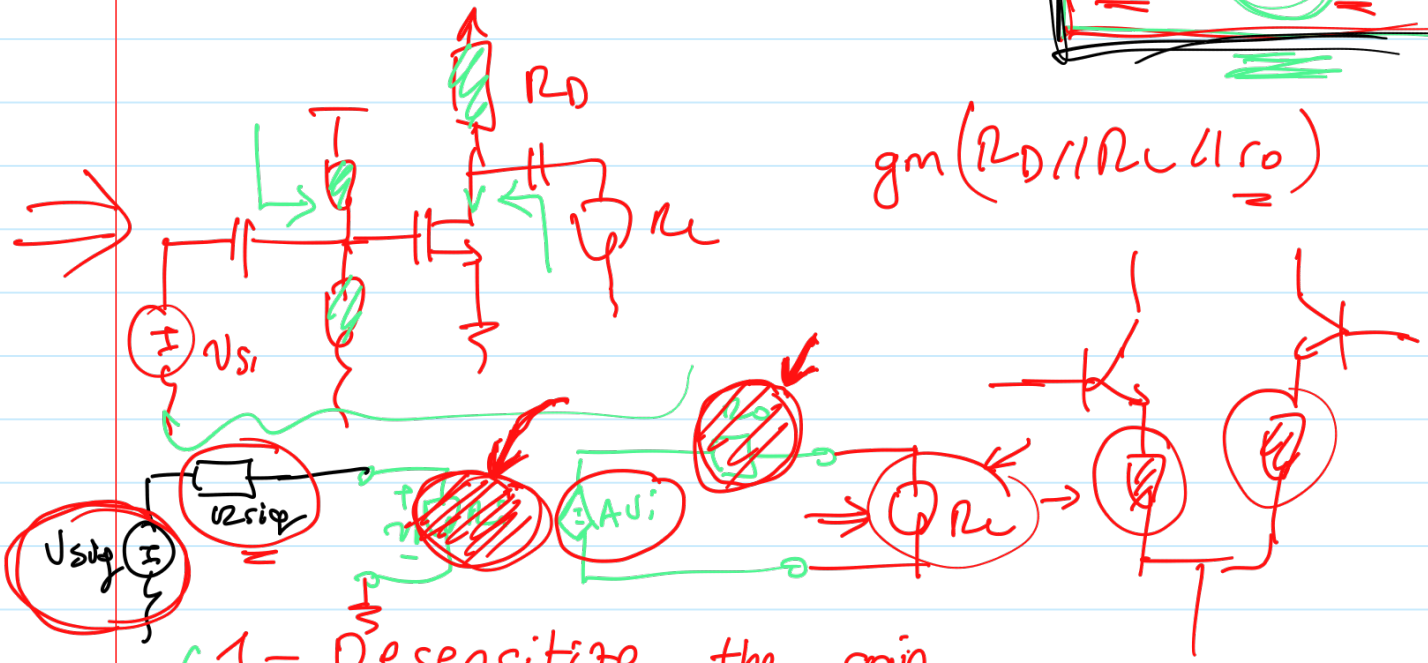


Feedback in OPAMP Circuits



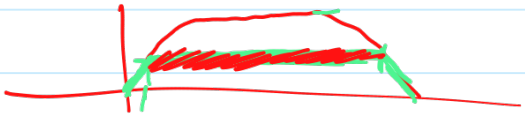
$$\frac{v_o}{10} \qquad \frac{v_n}{1k} = \frac{v_o - v_i}{9k}$$

$$9\,v_n = v_o - v_i \Rightarrow \boxed{v_o = 10\,v_i}$$

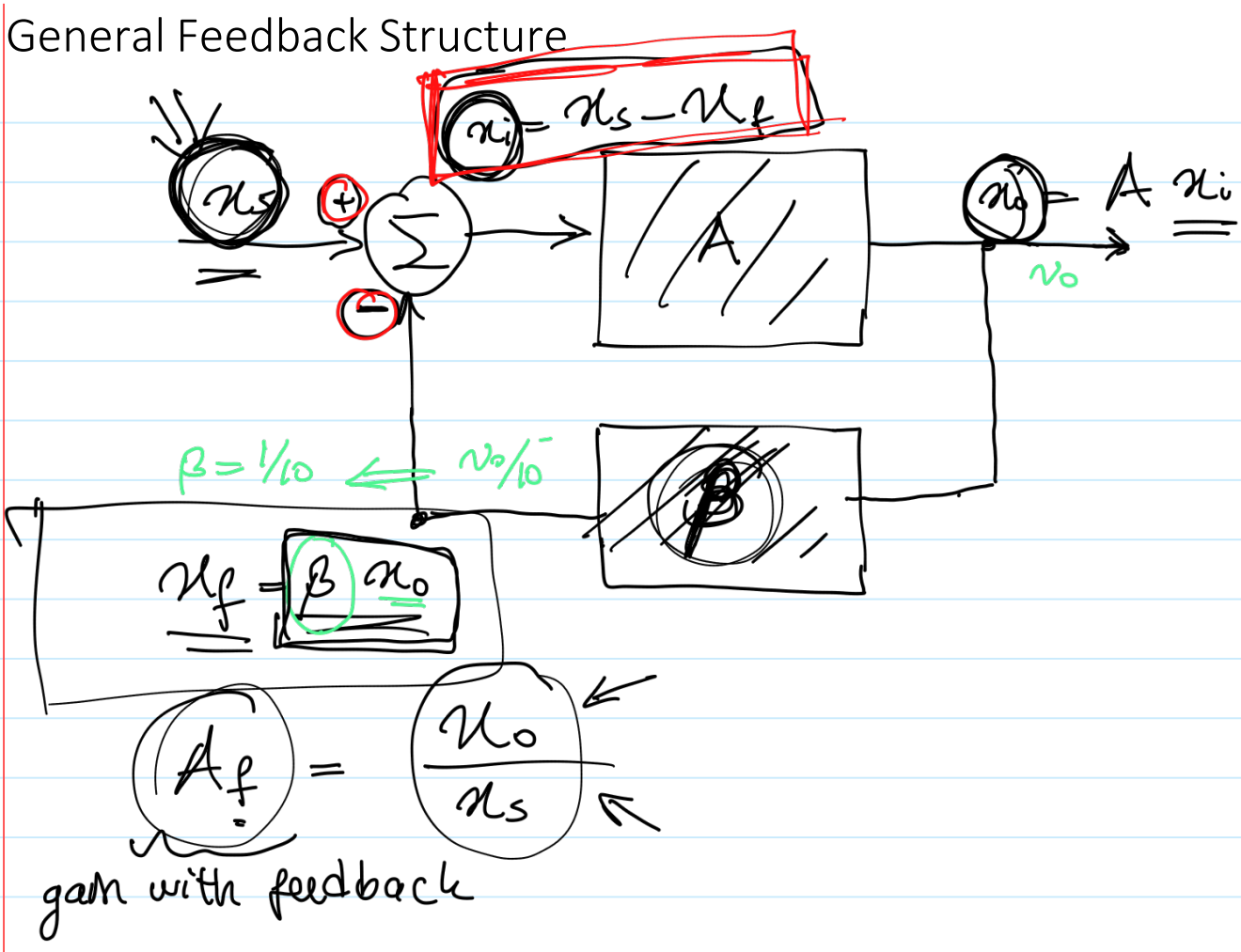


$$g_m(R_D \| R_L \| r_o)$$

1- Desensitize the gain
2- Reduce Nonlinear Distortion
3- Reduce the effect of noise
4- Control the input and output resistances
5- Extend the bandwidth



General Feedback Structure



$$x_o = A.(x_s - x_o\beta)$$

$$x_o = Ax_s - A\beta x_o$$

$$x_o(1+A\beta) = A x_s \Rightarrow \boxed{\frac{x_o}{x_s} = \frac{A}{1+A\beta} = A_f}$$

$A\beta \gg 1$

$$A_f = \frac{A}{1+A\beta} \rightarrow \frac{1}{\beta} = \frac{1}{1/10} = 10$$

$$\boxed{A_f\Big|_{ideal} = \frac{1}{\beta}}$$

$$A_f = \left(\frac{1}{\beta}\right)\frac{A\beta}{1+A\beta}$$

$$A_f = A_{f|_{ideal}} \cdot \left(\frac{1}{1+(1/A\beta)}\right)$$

deviation from ideal value

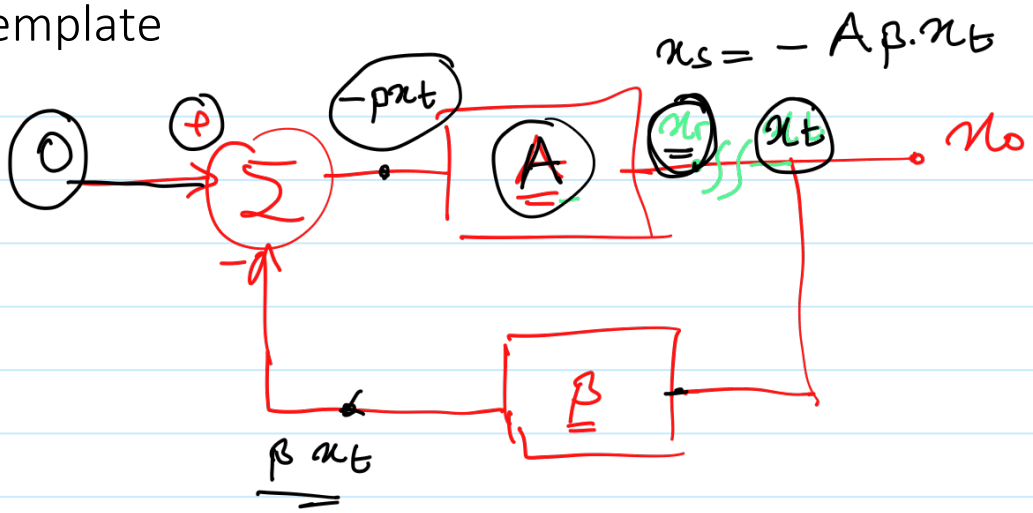
$$x_f = \beta \cdot x_o = \beta \cdot \frac{A}{1+A\beta} \cdot x_s$$

$$x_f = \frac{A\beta}{1+A\beta} x_s$$

$$x_i = \frac{1}{1+A\beta} x_s$$

$A \gg 1/\beta$

$$x_i = \frac{1}{A\beta} x_s \approx 0$$



$$x_s = -A\beta . x_t$$

$$\boxed{-\frac{x_s}{x_t} = A\beta}$$

$$\lim_{A \to \infty} \Longrightarrow \quad x_i = 0$$

$$x_f = x_s$$

$$x_o = \frac{1}{\beta} x_s$$

Example

Summary

Table 11.1 Summary of the Parameters and Formulas for the Ideal Feedback-Amplifier Structure of Fig. 11.1

- Open-loop gain $\equiv A$
- Feedback factor $\equiv \beta$
- Loop gain $\equiv A\beta$ (positive number)
- Amount of feedback $\equiv 1 + A\beta$
- Closed-loop gain $\equiv A_f = \dfrac{x_o}{x_s} = \dfrac{A}{1 + A\beta}$
- Feedback signal $\equiv x_f = \dfrac{A\beta}{1 + A\beta} x_s$
- Input signal to basic amplifier $\equiv x_i = \dfrac{1}{1 + A\beta} x_s$
- $A_f \mid_{\text{ideal}} = \dfrac{1}{\beta}$
- Closed-loop gain as a function of the ideal value $\dfrac{1}{\beta}$: $A_f = \left(\dfrac{1}{\beta}\right) \dfrac{1}{1 + 1/A\beta}$
- For $A = \infty$, $x_i = 0$, $x_f = x_s$, $x_o = \dfrac{1}{\beta} x_s$, $A_f = A_f \mid_{\text{ideal}} = \dfrac{1}{\beta}$
- For large loop gain, $A\beta \gg 1$,

  $A_f \simeq \dfrac{1}{\beta}$ $\quad x_f \simeq x_s$ $\quad x_i \simeq 0$

Gain Desensitivity

Bandwidth Extension

Reduction in Nonlinear Distortion

Template